FSF3827 Assignment 4

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Question 1

Part 1

Region defined by L_1 Norm Inequalities

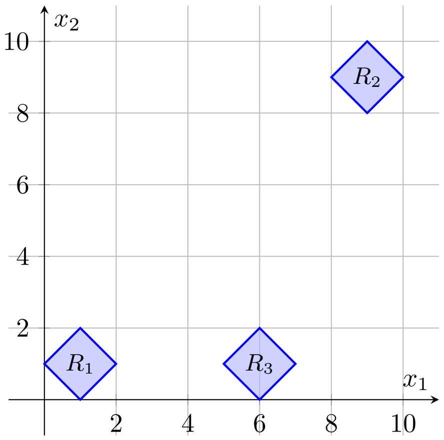

Figure 1: Visualizing the regions

Part 2

To construct the Big-M formulation, we first linearize the 1-norm constraints. The condition $\|x-C\|_1 \leq 1$ can be expressed linearly by introducing auxiliary variables z_i such that $\sum z_i \leq 1$ and $|x_i - C_i| \leq z_i$.

Let $b_1, b_2, b_3 \in \{0, 1\}$ be binary variables indicating active regions R_1, R_2, R_3. We introduce auxiliary variables $z_{k,i} \geq 0$ for each region $k \in \{1, 2, 3\}$ and dimension $i \in \{1, 2, 3\}$.

The Big-M formulation with the smallest valid single M values is derived as follows:

- **Region 1 ($b_1 = 1$):** Center $C_1 = (1, 1, 1)$. The maximum distance to a point in R_2 is 25, and to R_3 is 7. Thus, $M_1 = \max(24, 6) = 24$.
- **Region 2 ($b_2 = 1$):** Center $C_2 = (9, 9, 9)$. The maximum distance to R_1 is 25, and to R_3 is 19. Thus, $M_2 = \max(24, 18) = 24$.
- **Region 3 ($b_3 = 1$):** Center $C_3 = (6, 1, 2)$. The maximum distance to R_1 is 7, and to R_2 is 19. Thus, $M_3 = \max(6, 18) = 18$.

The formulation is:

Region 1:

$$\sum_{i=1}^{3} z_{1,i} \leq 1 + 24(1 - b_1)$$

$$-z_{1,i} \leq x_i - 1 \leq z_{1,i} \quad \forall i \in \{1, 2, 3\}$$

Region 2:

$$\sum_{i=1}^{3} z_{2,i} \leq 1 + 24(1 - b_2)$$

$$-z_{2,i} \leq x_i - 9 \leq z_{2,i} \quad \forall i \in \{1, 2, 3\}$$

Region 3:

$$\sum_{i=1}^{3} z_{3,i} \leq 1 + 18(1 - b_3)$$

$$-z_{3,1} \leq x_1 - 6 \leq z_{3,1}$$

$$-z_{3,2} \leq x_2 - 1 \leq z_{3,2}$$

$$-z_{3,3} \leq x_3 - 2 \leq z_{3,3}$$

$$b_1 + b_2 + b_3 = 1$$

$$0 \leq x_1, x_2, x_3 \leq 10$$

$$z_{k,i} \geq 0, \quad b_k \in \{0, 1\}$$

Part 3: Improved (Lifted) Big-M Formulation

To strengthen the continuous relaxation, we apply the lifting procedure to calculate specific Big-M coefficients for each region, rather than using a single global worst-case value.

Let us define the disjunctive constraints for region R_k as $g_k(x) \leq 0$. In the standard Big-M, we used $g_k(x) \leq M(1 - y_k)$. This is equivalent to $g_k(x) \leq M \sum_{j \neq k} y_j$. The improved formulation replaces this with:

$$g_k(x) \leq \sum_{j=1}^{3} M_{k,j} y_j$$

where $M_{k,j}$ represents the amount by which the constraint defining Region k needs to be relaxed to be valid in Region j.

1. Calculation of Coefficients

We calculate $M_{k,j} = \max_{x \in R_j}(g_k(x))$. Since the regions are defined by L_1 distances from centers C_k, the value $M_{k,j}$ corresponds to the L_1 distance between the center of region k (C_k) and region j (C_j), as the radius of the target region is 1 and the constraint allows a radius of 1.

- **Constraints for Region 1 ($C_1 = (1, 1, 1)$):**
  - $M_{1,1} = 0$ (Valid in own region).
  - $M_{1,2} = \|C_1 - C_2\|_1 = 24$.
  - $M_{1,3} = \|C_1 - C_3\|_1 = 6$.
- **Constraints for Region 2 ($C_2 = (9, 9, 9)$):**
  - $M_{2,1} = \|C_2 - C_1\|_1 = 24$.
  - $M_{2,2} = 0$.
  - $M_{2,3} = \|C_2 - C_3\|_1 = 18$.

- **Constraints for Region 3 ($C_3 = (6, 1, 2)$):**
  - $M_{3,1} = \|C_3 - C_1\|_1 = 6.$
  - $M_{3,2} = \|C_3 - C_2\|_1 = 18.$
  - $M_{3,3} = 0.$

2. Resulting Formulation

Using the linearized variables $z_{k,i}$ from Part 2, the improved constraints are:

$$
\begin{aligned}
&\text{Constraints defining } R_1 : \\
&\sum_{i=1}^{3} z_{1,i} \leq 1 + \mathbf{0}y_1 + \mathbf{24}y_2 + \mathbf{6}y_3 \\
&- z_{1,i} \leq x_i - 1 \leq z_{1,i} \quad \forall i \\
\\
&\text{Constraints defining } R_2 : \\
&\sum_{i=1}^{3} z_{2,i} \leq 1 + \mathbf{24}y_1 + \mathbf{0}y_2 + \mathbf{18}y_3 \\
&- z_{2,i} \leq x_i - 9 \leq z_{2,i} \quad \forall i \\
\\
&\text{Constraints defining } R_3 : \\
&\sum_{i=1}^{3} z_{3,i} \leq 1 + \mathbf{6}y_1 + \mathbf{18}y_2 + \mathbf{0}y_3 \\
&- z_{3,i} \leq x_i - C_{3,i} \leq z_{3,i} \quad \forall i
\end{aligned}
$$

3. Discussion and Analysis

Motivation for Strength:
The improved formulation is stronger because it uses tighter coefficients. In the standard Big-M formulation for Region 1, the term was $24(1-y_1) = 24y_2+24y_3$. The improved formulation uses $24y_2+6y_3$. Whenever the continuous relaxation assigns a non-zero value to y_3 (e.g., $y = (0, 0.5, 0.5)$), the RHS of the improved constraint is smaller ($24(0.5)+6(0.5) = 15$) than the standard formulation ($24(0.5)+24(0.5) = 24$), effectively cutting off more infeasible fractional solutions.

Coefficient for the own variable (M_i):
We do not need to compute the coefficient M_i for the binary variable y_i associated with the constraint's own region. It is always **0**. This is because the constraint must be satisfied exactly when $y_i = 1$, meaning no relaxation (slack) is allowed.

Equivalence to Conforti Section 7.2:
Yes, the resulting inequalities are equivalent to the lifting procedure described in Section 7.2 of "Integer Programming" by Conforti et al.. In that framework, lifting involves starting with an inequality valid for a restriction (e.g., $y_1 = 1, y_2 = 0, y_3 = 0$) and computing the tightest possible coefficients for the variables fixed to 0 (y_2, y_3) to make the inequality globally valid. Solving the optimization problem $M_j = \max(a^\top x - b)$ s.t. $x \in R_j$ is exactly the separation problem required to determine these lifting coefficients.